

Data Primer: Making Digital Humanities Research Data Public

Felicity Tayler, Marjorie Mitchell, Chantal Ripp, Pascale Dangoisse

Collaborators:

Constance Crompton (University of Ottawa)

Karis Shearer (University of British Columbia)

Matthew Lincoln (Carnegie Mellon University)

Mikhel Proulx (Concordia University)

Alex Gregory

Alex Wermer-Colan

Alexei Alexeev

Alicia Cappello

Andrea Fatona

Andreea Ritivoi

Augustine Farinola

Beth Knazook

Cynthia Heider

Deanna Reder

Ece Turnator

Erin Galt

Evonne Levy

Fatoumata Bah

Federico Pianzola

Katelin Parsons

Lauren Churilla

Lisa Goddard

Lorna Dawes

Maria Laura Flores Barba

Marie Noël

Martha Attridge Bufton

Megan Toye

Monique Grenier

Nadine Valcin

Paul Albert

Sally Hadden

Samuel Mickelson

Shahira Khair

Stefan Higgins

Yann Ryan

Victoria Sciancalepore

This Primer was collaboratively authored by over 30 Digital Humanities researchers, research assistants and data professionals and reflects the disciplinary-specific RDM instruction method developed for the SSHRC-funded partnership, the Spoken Web. Participants consented to having their authorship acknowledged as part of the collaborative document authoring process, and were informed of their inclusion as authors again before

publication. We gratefully acknowledge our open peer-reviewers Jane Fry and Donna Langille for their commentary.

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Who is this Primer for?	• The best practices outlined here can be applied across the broad spectrum of digital humanities (DH) methodologies. • This Primer will be most useful to experienced humanist researchers, working in Canada or with Canadian funding or research affiliation, who wish to improve their data curation strategies for collecting, managing, and publishing digital files and formats alongside traditional textual scholarship. • The focus of this guide is collections-based methodologies that make a distinction between research materials such as physical archives that may not belong to the researcher; and the research data that the researcher may produce and publish.
How does this Primer work?	• This Primer is an overview of the different aspects of data curation and management best practices. Its intention is to get you started on thinking about how to integrate best practices into your projects. It is not a step-by-step guide, but we do provide you with useful, technical guides in the appendix.

	• This Primer outlines how the Data Flow and Discovery Model helps digital humanists assess and plan their data curation and management needs as an iterative process that can be conducted throughout the life of their research project. • This Primer is researcher-centred; each section begins with questions that were raised in workshop breakout groups. Those questions were then mapped onto relevant sections of the data work-flow tool to create a conceptual model that follows best practices in data management. It is important to note, however, that while the Primer is researcher centered, the researcher is just one actor in the flow of data; they are not the sole decision taker, nor the central one, when it comes to data curation planning. • The SSHRC-funded SpokenWeb partnership is used as an example throughout the Primer to illustrate how best practices were followed and incorporated into a data curation project. Case study examples will appear in separate sections throughout the Primer to illustrate key data curation strategies.
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Overarching principle	We acknowledge that this Primer is written from the perspective of settler librarians and academics. We also acknowledge the importance of Indigenous data sovereignty, and strive to develop and abide by principles that recognize and respect First Nations, Inuit and Métis People's rights of data sovereignty, self-determination, and self-government through a distinctions-based approach (Taken from the Tri-Agency Research Data Management Policy – whose policies should not supersede First Nation, Métis and Inuit laws, protocols, and policies). We also acknowledge that digital humanities and data curation management methods have had their roots in oppressive practices or frameworks towards other historically excluded or marginalized communities. Again, we strive to guide researchers towards using non-oppressive approaches to data curation.
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What is data curation for digital humanists?

Data curation may have different meanings depending on your field of study, country, or organizational culture. For the purpose of this Primer, we use the [Portage Network](#) view that data curation is the active collecting of research data as it is created, maintained, used, archived, shared, and reused. Data curation is an iterative process with the ultimate goal of creating datasets that are [FAIR](#) (Findable, Accessible, Interoperable, and Reusable). Datasets with Indigenous ownership or origins should respect the [CARE](#)

[principles](#) (Collective Benefit, Authority to Control, Responsibility, and Ethics) and any domestic principles established by First Nations, Métis, or Inuit Peoples in Canada (for example the First Nations principles of OCAP[®] (Ownership, Control, Access, and Possession)).

Research data curation management consists of multiple processes that are applied to your research project throughout its lifecycle. It includes collecting, documenting, storing, sharing, and preserving or the destruction or return of research data—now and in the future. Data curation is one very important process within the steps of research data management [[Portage Primer: RDM](#)].

Data management and curation are important processes for digital humanists, because, without proper planning and management, the value of the data as well as the labour involved in researching, collecting, and analyzing the data, could be lost either while the project is underway, or after, once it is sent off for preservation and sharing. An added benefit of data curation and management is that it makes your data easier for both you and your project collaborators to work with. Further, most funding agencies (including the Tri-Agency) require data management plans and data deposits that render research data FAIR ([Portage network, curation expert group](#)). If you are working with data that has Indigenous ownership and origins, data curation is a crucial step towards respecting conditions of data governance.

When do you start data management planning for curation?

Data curation management should be integrated from the very beginning of a project, during the planning stages. It should be a core part of the methodology of your research project. Reviewing your data management requirements during the planning stages of your project also allows you to reach out for assistance and ensure your project team has the necessary skills before you experience any problems or roadblocks. Data humanists tend to focus on data curation management once a research project is finished, either when it's time to analyze the data or when it's time to decide where the data should be stored. Don't do this.

During the planning stages, consult with research management data support services, which may include librarians and advanced research computing experts. Looking at existing peer digital humanities research projects in the same discipline or using similar methodologies also ensures that you build on existing community practices for creating, formatting, and publishing your data. It is also very important to consult with your ethics board and/or copyright office during the planning stages, as their expertise may also be required. Collaborating with a librarian ensures you have the necessary expertise in data management best practices and repository options. Collaborating with experts in advanced research computing, either through your institution or a national organization, ensures you have the necessary technical skills and infrastructure you require for your project.

Research conducted by and with First Nations, Métis and Inuit communities, collectives and organizations, may require you to go beyond the requirements of the [Tri-Council TCPS2](#), the [Tri-Council RDM policy](#), and your institutional ethics board. Working with respect for data sovereignty means that First Nations, Métis and Inuit Nations, communities, collectives, or organizations should guide and ultimately determine how their data are collected, used, and preserved, and make sure that they will have the right to repatriate their data. Planning your data curation management at the onset of the project will better integrate the [First Nations principles of OCAP[®]](#), Métis principles of OCAS, and/or the [National Inuit Strategy on Research](#), or the broader, global [CARE principles for Indigenous Data](#)

[Governance](#), and other relevant non-oppressive approaches to your data collection, protection, sharing, and use.

Be aware of First Nations, Métis and Inuit communities' data curation capacity needs and, in consultation with your partners, consider ways of supporting such capacity development in your proposed work from the outset, e.g., funding applications, your own institution's data infrastructure planning, etc.

Thorough planning also allows for working out what your final outputs across media platforms may be: traditional publications (monograph, article), data sharing, knowledge mobilization through non-traditional digital formats, or data papers.

The SpokenWeb Partnership is comprised of 13 institutions across Canada and the United States. It develops collaborative approaches to literary historical study, digital development, and critical and pedagogical engagement with diverse collections of literary sound recordings from Canada and beyond. The SpokenWeb is a significant producer of humanities data, derived from scholarly and creative work with audio recordings. This includes :

- Analogue recordings from an historical period ranging from the 1950s to the 1990s;
- Born digital recordings that document oral history interviews and events taking place in the present day;
- Many types of files produced at each partnership site: audio; administration forms and documentation; spreadsheets; software and web code; and
- Images files and all derivatives from activities such as research creation, pedagogical tools, digital storytelling, and scholarly writing, etc.

From the beginning, the SpokenWeb partnership has understood data curation to be a core part of the methodology of its research project. In other words, the listening practices that are the basis of forming community around aural experience in the partnership are paralleled by the enabling of the research community through data sharing.

Arguably, the key to successful data curation and sharing across the SpokenWeb partnership has been a requirement that every partner institution bring on a librarian co-applicant. This requirement for disciplinary researchers to pair up with stewards of library and archives collections and information systems, provides the necessary expertise in data management best practices as established by a national network of RDM practitioners, across the partnership.

How do you work through the data curation management process?

The Data Flow and Discovery Model (see the annex for an example worksheet) is a way of representing the flow of your data through digital humanities methodologies; it gathers information about the material and immaterial inputs and outputs of each entity and the process itself. Specific operations performed on the research material and data can be represented by a flowchart.

Although a flowchart suggests progress in one direction, the steps are interrelated. Consideration of final storage, for instance, may impact the earliest stages of data collecting. Likewise, you will notice

that some of the crucial choices you make in how you organize and store your files have an impact on your ability to transform your data into scholarly or creative work, or your ability to govern conditional access.

The following section delves deeper into how to use a Data Flow and Discovery Model, section by section. Remember that even if this Primer calls for a linear representation of the model, it's important to note that it is an iterative, sometimes repetitive, process. In many instances, some steps will have to be reviewed or skipped over and looked at in a different moment in time, depending on your project data, goals, and progress. Sometimes the most important outcome of this modeling is noting "further questions", if there are steps in data curation management where you don't know what to do.

Consent and intellectual property



Data Flow Model: Consent

Researcher question(s): I have different types of consent, and layers of ownership to negotiate, where do I start? What if you have to go back and retrace ownership for publishing permissions? What consent and permissions should you seek out early in a project? What could I do differently going forward? Are my artworks and other forms of cultural expression "research data"? What permissions apply to the owned works versus the data I create in studying them?

Data Flow model questions that help you think through this complexity: Who is the owner of the data? Are there ownership or licensing conditions that will shape access and reuse of your data?

At the start of the project, it is important to [understand the distinction between research material, and research data](#) as they are managed differently. Research material is the source, the impulse for your inquiry and it is unprocessed; whereas research data is processed and is often more accessible than original research material. According to the Tri-Council, research material is the 'object of investigation' (think of material objects, such as archival sources or items in a museum collection), whereas the research data is created through your methodologies as you engage with your research material (such as your findings or results). However, when working with a principle of Indigenous data sovereignty (with information from or about First Nations, for example) this distinction matters less. What matters most is the roles and relationships towards the information; researchers per se may not have a say in how "research materials" were collected or are managed and governed, but archivists do. For example, from a First Nations perspective neither the research "material" or research "data" should be released without First Nations permission.

You should make this distinction between research material and research data as the consent or licensing conditions for access may be different. You may not have the license to re-share the research material you are studying, but you may have the right to create data and share this data as part of your larger research project output. Your rights to share your research data will depend on your collaborators as well, and negotiation for what will be shared, and how, also needs to be done in the early stages of the project.

Choosing how open your project is does not solely rely on your wishes and ideals. The use and reuse of data for publishing is contingent on contextual conditions of consent, ownership, and permissions. Copyright and informed consent must be tracked to reuse the data. When accessing your different types

of consent, permissions, and rights, you can think that the end goal is to make the data as open as possible but not necessarily open. Or “as open as possible and as closed as necessary” [\[FAIR\]](#). That is to say, various layers of consent and permissions will shape how open the data you wish to reuse and publish *can* be. Your desired use of the data may not determine what is possible for your project, but rather the constraints of your informed consent protocols or intellectual property licenses, or if you are working with First Nations, Métis, and/or Inuit research partners who have ownership of their data. Plan your data curation early in the process; it allows you to potentially partner up with data owners at the onset of the project and move along the data flow process in partnership with other actors, instead of finding yourself hitting an impasse during the process.

Note: The technical details of how these principles will apply in your specific research project are beyond the scope of this guide. It will be up to you to define and find the resources to support your specific technical requirements.

Research material for the SpokenWeb project consists of the original audio recordings, and the research data is the metadata describing these recordings, or interpretation and analysis of the content of the recordings. For example, researchers use the audio recordings (research material) to do a gender analysis of how much time is allocated to women or men during a literary reading event (research data). Or, researchers use the audio recordings (research material) to create metadata descriptions listing authors’ names audible on the recording (research data).

Further, SpokenWeb has digitized analogue “literary” events recordings, born digital “literary” event recordings, and born digital recordings (oral history). The original analogue recordings and their cases are “research materials” and the digitized versions and their metadata descriptions are “research data.” (For more details about metadata, see the Data Collection section of this guide.)

Copyright

Copyright protects original expressions of content that are fixed in a tangible medium (either physical or digital) – note that there are some differences between various national copyright regimes, and in most cases the law applies in the national space where the expression is created. We recommend following up with your own institutional library or copyright office for specific questions about licensing.

Data could be protected by copyright if there was judgment or skill applied in the choice of data to include or its arrangement (raw data is not intellectual property, it is facts in public domain). External factors to your research team or project may determine if data is protected by copyright and who owns it. It is advisable to document data and copyright ownership in the early stages of a project.

A licence is a permission provided by the copyright owner allowing someone else to use their work for certain purposes and under certain conditions. The copyright remains with the copyright owner.

Open licences are used by copyright owners to indicate which rights they wish to keep, while communicating which types of use others can make of their work, without having to ask for permission every time.

Some examples of open licences*

<u>Licence</u>	Distribution	Modification	Sublicensing	Attribution
Copyright	Ask permission	Ask permission	Ask permission	Required
CC BY	Allowed	Allowed	Allowed	Required
CC0	Allowed	Allowed	Not allowed	Not required
ODbL	Allowed	Allowed	Not allowed	Required
ODC-BY	Allowed	Allowed	Not allowed	Required
PDDL	Allowed	Allowed	Allowed	Not required
MIT	Allowed	Allowed	Allowed	Required
GNU GPL	Allowed	Allowed	Allowed	Required
Apache	Allowed	Allowed	Allowed	Required

* All 8 open licences allow for commercial use

Sublicensing = derivatives can be shared under a different licence

Source: Mélanie Brunet & Thomas Rouleau, Copyright and Research Data at uOttawa – FAQ (2021) under CC BY

The above table shows us, from the point of view of a researcher who wishes to reuse content that is not under their ownership, how copyright compares to popular open licences for data sharing or for sharing software or code. You may apply different licenses to the published research, writing, data, and the code outputs of your research. Note that, because the particular mechanics of code and data reuse differ from the way “creative works” (including academic writing) are reused or re-published, it is often recommended to use code and data-specific open licenses, instead of applying the same creative commons license that you may use for the creative output of the project ([Creative Commons](#)).

When planning for your own publication outputs, you want to make distinctions between open licensing and licensing that limits use. For example, you probably want to keep copyright on your research articles, and you want to be sure that you respect the copyright and licensing agreements with people whose intellectual or creative content you are sharing through various knowledge mobilization activities; however, you may want to put an open data licence on your metadata export.

Research material and research data is always situated from the perspective of the researcher. If you are working in research-creation and your outputs are artistic or cultural expressions that have value beyond academia, you may want to consider that your “research data,” as it is produced, circulated, cited and reused in an academic economy, is nonetheless creative work that belongs to you. Perhaps it should remain in copyright and not be shared through data publishing platforms. Or, you could consider your own art as possible potential research material for future researchers and artists. , It could be advantageous for you and the research community to take time to learn how to consciously use open licences and data repositories as parallel self-publishing and self-archiving systems that allow you to share your artistic expression on your own terms.

First Nations, Métis and Inuit laws on intellectual property including cultural expression and traditional knowledge, may be different from national and international copyright and intellectual property law. Note, for example, that the Canadian Copyright Act awards copyright in stories, songs, poetry, etc., to the first person to record them—that is, capture them in print or audio recordings. This has allowed traditional stories or songs to be ‘owned’ by researchers, rather than the Indigenous peoples who created them. This law does not respect the oral traditions of many Indigenous peoples in Canada.

When working with data with Indigenous ownership, it is essential to inform yourself about Indigenous laws and their implications to your work. Each context of engagement will be different and shaped by distinct protocols of First Nations, Inuit, and Métis communities.

In the SpokenWeb partnership, common standards that enable data sharing among the partnership and eventually, to the public include

- *a shared ethics protocol with informed consent that permits data sharing to the public, and*
- *a shared digitization waiver for historical audio artefacts.*

The primary site PI (Jason Camlot) sets a standard that other partner site PIs then follow. An Ethics protocol form for oral history interviews was submitted through Concordia University. Once this protocol form was accepted by the Concordia University Human Research Ethics Committee, it could then be adjusted to fit the requirements of the partnership site REBs and submitted for expedited review.

This means that the privacy and confidentiality restrictions on the use and sharing of these interviews are standardized across the partnership. These restrictions are set according to the interviewees choices made on the Consent Form. These interviews are “data” that have specific attributes regulating sharing. The principles of these ethics protocols for oral history interviews are also applied to public events held by the SpokenWeb partnership. For example, Symposium attendees fill out a permissions and waiver form for recordings and future publications of their sessions.

Data Flow Model: Data Collection

Researcher questions: If I am working with an anti-oppression framework or decolonial methodologies, should I use an existing database or descriptive system or invent a new one? What standards exist already? What are the labour, resources, relationships needed to adapt it? Is my process of adaptation or imagining new modes of description part of my data? Do I need to capture it, and if so, which parts of the process?

Data Flow model questions that help you think through this complexity: Can you list your data types? What files will need metadata? Do you have protocols for collective ownership?

Start by thinking about all the different formats and types of data you will gather and create during your research project. Also think about where they reside and how they are used in the project. Ensure to organize them and to be consistent in your collecting processes and in your descriptive standards. Create an inventory of your data types and what file formats they are stored in. This step will also help you choose what is within or out of scope in terms of processing, preservation, and sharing.

As you organize your data and files, decide on naming conventions, and start to think about how to structure folders and files. Naming conventions (i.e., using a consistent and descriptive way to name your files) are important to easily locate the right files and to help with version control. This can seem tedious initially, but the frustration of opening a series of different files to try to find the correct document or version can be avoided by following simple guidelines. To structure your files, use folders (that also adhere to a [naming convention!](#)), be precise in what goes in each folder, and organize them hierarchically (from broader topic to more specific, i.e., Administration > Communications > Emails >

Emails from Felicity). If the data is being returned or provided back to a First Nations, Métis or Inuit community, this structure should also be built in a way that reflects their needs and priorities, and is accessible to them and meets their capacity.

Plan your data collection with consideration of the contribution that the newly collected data may or may not make to existing data resources. Whether this contribution happens at the same time that you collect your data or later on in the project, this plan to merge your data with a pre-existing system will impact how you gather and record your data.

Best practices for managing your data include making a distinction between:

- Your administrative data (RA contracts, project management files, budgets, etc.) and your research data
- Your research materials and your research data
- Your platforms, software, code, and other resources developed to share or visualize your research
- The content of the data and the file format in which the data is stored (for example, it is possible to store texts in spreadsheets or databases in addition to storing texts in word documents).

The technical details of how you will plan and carry out your data collection for your specific research project are beyond the scope of this guide. It will be up to you to define and find the resources to support your specific technical requirements. However, once you've figured out your data collection plan, file structure, and naming conventions, it's time to get started on adding context to your data to ensure others understand it. That's where metadata comes in handy.

Metadata is the data that provides information about the data. Metadata is the best way to describe your data and to point back to the original sources; at this point, it's a good idea to start thinking about the metadata that will be needed for your project files.

Do you remember when we made the distinction between research material and research data in the first section? In that case, the descriptions of the recording, and lots of the SpokenWeb partnership research data is actually metadata.

SpokenWeb began by researching and adopting existing metadata schemas for literary collections and audio file descriptions. This saved time, and also ensured that the data they produced could be imported into other existing systems, such as library catalogues and Wikidata. It also meant they could import or borrow metadata from existing systems. But the SpokenWeb teams also noted limitations that these schemas had, and built the necessary technical skills and labour into their project methodology in order to develop custom Metadata and Cataloguing conventions for analog audio artefacts. All audio artefacts are being described in a metadata ingest system affectionately named SWALLOW – alluding to the songbird of Romantic poetry and to the “swallowing” or ingesting of the data -- this means that all collaborators in the partnership can search each other’s audio collections. Also, SpokenWeb Research Assistants are all trained in metadata entry and represent the majority of labour deployed in this contribution to the scholarly project.

Use metadata to describe your data by type, use, title, source, etc. for different audiences; it is also a way to track licensing, and conditions of consent. Metadata must be machine readable for computational searches, but also descriptive for humans (i.e., other researchers, collaborators, etc.) to allow them to find the data efficiently. Metadata can also be used to describe your methods. The methodology of your project is also data, and necessary information for collaborators, partners, and future researchers to cite, use or reuse your project.

Some disciplinary communities already have established metadata schemas (many can be found at the [Metadata Standards Catalog](#)), it is a good practice to start from there. By using an already existing schema, you can focus on modifying or expanding it to suit your needs, values, and questions. You then also contribute to the larger community's data potential, and this practice does fit into the FAIR principles of findability, accessibility, interoperability, and reusability. Catalogues of metadata schemas can be helpful, or you can consult your librarian and other institutional metadata specialist.

At a minimum, create a README file that explains the choices you have made in this metadata description, and it should also describe the file structures to facilitate the understanding of the collection or decisions and observations, so that your collection can be reusable. Each data type, or file may have several different types of metadata:

- Administrative (chains of commands, logs, manifests)
- Technical (properties, characteristics of file)
- Descriptive (summaries, annotations, context, key terms)
- Preservation (condition and quality of file for example), and
- Consent and permission focus.

You can treat the writing of metadata in different ways; if you have access to the technical skill or knowledge, you can adapt or build a cataloguing system for this metadata that will allow for sophisticated data imports, exports, linking to sources, and searching that goes beyond a spreadsheet; but at its simplest expression, you can also treat the metadata as if you were writing a text. Metadata can be viewed as an encoded narrative that tells the story about your data. The metadata can function as a layer of description that guides the interpretation of your project. Adding metadata can be as simple or as complicated as you wish it to be; the important thing to remember is to describe your data so that others can collaborate more effectively with you, reuse your data or repeat your methodology. If you find the previously created directory does not suit your work, you can see if there are ways to add fields to correct it or to acknowledge a reality that wasn't considered previously. In any case, it's always best to try to reuse than to start from scratch.

The SpokenWeb partnership has had conversations with Cree-Métis literary critic, Deanna Reder, about Indigenous research methods and protocols for literary scholars. Reder and the team members of [The People and the Text \(TPatT\)](#) rely on the First Nations Principles of OCAP[®]: ownership, control, access, possession (in Health, it is the 4 R's: respect, reciprocity, relevance, and responsibility) as well as nation or tribal-specific protocols. TPatT is inspired by work by Indigenous scholars and creatives such as Skawennati, Jason Lewis, Elizabeth LaPensee, and Maize Longboat, and benefits from the support and infrastructure of CWRC (Canadian Writing Research Collaboratory).

Data collection and ownership

When undergoing the data collection phase, DH researchers often engage in a reflexive testing of culturally imposed power imbalances in digital systems. This critical position presents unique challenges to RDM as a set of commonly adopted practices. At the same time, the bias towards principles of Open Scholarship makes DH an excellent gestation ground for best practices of culturally sensitive, cross-disciplinary, globally oriented data collecting and sharing.

There are many examples of self-defined communities that have their own tacit or explicit community or collective ownership protocols that impact your data collection process. All these approaches require the long-term development of community relations and may take timelines that surpass your funding deadlines. In these cases, the requirements of Research Ethics Boards ([TCPS2](#)), or the parameters of intellectual property law, do not go far enough to establish ethical approaches to data collection and subsequent sharing.

A laser sharp example of this need to work with community protocols is the assertion of data sovereignty by First Nations, Métis, and Inuit peoples in solidarity with global movements for Indigenous data sovereignty. This movement asks researchers who are working with Indigenous communities or with data that has Indigenous ownership to recognize that existing personal and individual consent and data privacy protections do not respect *community privacy*, or the collective consent and data privacy protections, supported via community-controlled data infrastructure, that are essential to ethical Indigenous data practices. Additionally, this movement recognizes that the intellectual property laws of colonial nations are in competition with the protocols of Indigenous law, traditional knowledge, and cultural expression. For example, the CARE principals can provide guidance in this area, or when working with First Nations communities, the OCAP® principles apply.

All of this attention to protocols feeds into the data collection phase as you need to work with the community before collecting data to ensure their rights, heritage and practices are respected. As described by the First Nations Information Governance Centre, “First Nations have an interest in information that describes them. It does not matter whether information is collected or held by First Nations or held by another entity such as government or a university. First Nations assert the right to govern how the information will be used and disclosed, and what limitations or conditions must be placed on its use. The right of First Nations communities to own, control, access, and possess information about their peoples is fundamentally tied to self-determination and to the preservation and development of their culture.” (OCAP®)

The FAIR principles, as well as the DH, push for open data, more sharing and open licenses conflicts with the Indigenous rights to assert control and ownership of the collection and use of Indigenous knowledge. The Global Indigenous Data Alliance ([GIDA](#)) and the United Nations Declaration on the Rights of Indigenous Peoples ([UNDRIP](#)) bring the expressions of sovereignty that take place in this global and local context to bear on data management best practices, and whereas these tools can be a good place to learn about working with First Nations, Métis, or Inuit communities, each individual community will have their own specific protocols to work with.

Working with your data



Data Flow Model: Data Processing

Researcher questions: I have all this data, and I understand it, but how will my research assistant, community partners, or future scholars in my field understand it? How will they be able to work with my data?

Data Flow model questions that help you think through this complexity: File formats? Number and size of files? Storage & backup locations? How are you recording metadata?

The data processing phase of data curation management can be understood as the moment when you convert your data into a usable and desired form according to your research, preservation, and sharing goals. This is also when you actively convert, transfer, or transform data into machine readable formats (and ideally in a format that will be shareable amongst many projects, collaborators, and platforms!), and when you look at where there are gaps or misinformation in the data. You may find this phase to be quite iterative, or tedious, especially as it is often undertaken in combination with the data collecting phase: coming back to your data files to process your data as more information is discovered and collected at later stages of the project – we all do get excited about the analysis phase of the data gathered, but this phase will allow for better and greater analysis in the long run. Again, this step should be thought out and undertaken right at the start of the project; process your data as you collect it. The cost of processing all your data later on may be very high, especially in current research environments that deal with large data sets. Moreover, analyzing data that was not accurately processed could slow down the work or generate misleading, unreliable, or inaccurate findings.

When processing your data, document each step -- the decisions you make are important to communicate to those who use the files after you and keep in a README file, for the same reasons you documented and described your data collecting process. This can be understood as recording your thought process, or your sorting process (I.e.: I'll include this and exclude that). This is in sorts documenting your "intellectual labour" as opposed to your "intellectual product" (e.g., my book chapter). If you are accustomed to working in archives you can also think of this documentation as a "finding aid" that helps you communicate the provenance and order of the files according to the relationships that created them. This is another example that fits well into the 'thinking process' to allow others to understand and replicate. For example, if you edit an 'original' dataset (don't forget to keep the original master file!) to conform to a given research project, this must be documented (changing headers for census data variables (which comes as long sentences) in an Excel table to be machine readable in another platform. Additionally -- think about the boundaries of the research -- which research files are to be shared among the partners and what are individual research files to be managed by individual researchers? Documenting the decisions made about processing the data will likely influence future analysis that will happen.

It is also at this point that you start processing each file's metadata, that is to say, that you actively add the metadata to each file. Yes, here again you also need to add the metadata structures and decisions in the README file.

To take SpokenWeb (UBC Okanagan) as an example, the processing of analogue audio involves the creation of three levels of digital files:

- Master (.WAV),
- Master Access (.WAV) and

- Access (usually .MP3, derived from the Master Access).

The Master File captures all audio data to the standard of 96KH / 24bit and may, depending on the nature of the event recorded, include "background" noise, like chairs scraping the floor, or dish clinking. If an Access File (derived from the Master Access File) is cleaned and the background noise is removed, auditory information is being scrubbed from the record and these edits must be documented so users understand what versions they are working with. For some researchers, this background noise data might not be important, but in other cases, it will constitute the research focus.

Transforming data management into scholarly and creative work



Data Flow Model: Critical Analysis

Researcher questions: Over the many years of my project, how do I make sure I can find all the files, digitized images, and other digital things that I need to write my papers, make podcasts, make conference presentations, etc.? I want to share files with my community and research partners but don't want them to go public? My research assistants have left the project and I don't know how to access the files they worked on?

Data Flow model questions that help you think through this complexity: What is the anticipated use of the data in the research team -- is there a need for access to files between multi-institutional partnership collaborators? Does file storage balance security with file sharing across collaborators?

Once you have collected and processed the data, it's time to ensure everything can be easily accessed by the right organizations and people so that you can do your scholarly and creative work. Different files need to be accessed by different members of your research team or partnerships, and not all team members will access the files for the same purpose or from the same location. For many DH researchers working in a team, their "critical analysis" phase is dependent upon well-organized, and *accessible* files. The data needs to be curated, described, and made accessible for theories, histories, narratives, discourses, or dialectics to be gleaned.

Before considering a specific technical solution to sharing files, you must have a system-agnostic file management strategy which involves having:

- 1) File and directory naming conventions so that individual files can be recognized even if removed from their original context
- 2) Ensuring that references to filenames in data/metadata files are valid
- 3) Documenting the conventions in a README file (yes, the README file again!) so future project collaborators can learn the file management plan and have documentation to update as the project changes scope.

Also, consider what differences there may be between the research data that your team shares and processes for the duration of the project, versus the format of data that you ultimately share out as part of publication and preservation. Differences may include varying license and re-use restrictions, but also practical matters, such as size of data, format, and bandwidth requirements. You need to securely store, version control, and backup files, but also want a system that allows restricted access while RAs and researchers work on them. Also, confidentiality and copyright may restrict access to files in different ways. You must also look at the various sharing practices between partnerships or institutions: if data is located at one institution in a multi-institutional project, are there issues of access for team members that are not based at those institutions? You may need to collaborate with IT support teams in different locations – this takes time and planning!

In your larger research plan, take time to pause and assess whether the data management and sharing solutions you worked out at the start of the project still fit the scope, questions, and sources that your research project has evolved to encompass. It may be that you now have many more (or much larger) files to manage; your original file-naming convention may have assumed all your objects were coming from one institution, and now you are incorporating items from other institutions and need to now track those varied sources explicitly; you originally only worked with collaborators within your institution, and now you want to bring in external collaborators, etc.

How will you manage onboarding of new, incoming RAs and everyone else new to the research project? Take time to do an orientation of your research group, and have a common set of “guidelines” or “rules” about what is acceptable use of the research materials, and what isn’t. Answer common questions and “worst case scenarios”. For example, what if an RA thinks it is fine to copy research files to their personal laptop. Is that an acceptable practice? Who can they share those files with? Or not? Explicitly stating expectations allows all to participate with a shared understanding. Sometimes a project binder (a soft or hard version), created by the project lead and unique to each workflow, in which all project protocols and best practices are recorded, can be the most practical way to go about this.

Note: Research Assistants have unique opportunities on digital humanities projects to build technical skills and knowledge of metadata entry. These are not only excellent skill sets for them to graduate with, but it is also likely how a large percentage of the labour needs of a digital humanities project will be resourced. It is important to view this labour as a significant contribution to the scholarly work in future publications even if/after their contracts have expired. We recommend using a protocol such as the [Credit Taxonomy](#) or [The Student Collaborator’s Bill of Rights](#) to acknowledge your student labour in co-authorship or other appropriate roles; anonymous skill-building does not go far enough in helping them build their scholarly careers.

Concordia has solicited Compute Canada to join SpokenWeb as an institutional partner; as a partner, this national data infrastructure service provides cloud storage space and basic services to all SpokenWeb partners who choose not to use local infrastructure options. Other SpokenWeb partners, such as UBCO, have chosen to use their own local institutionally provided cloud storage infrastructure, but contribute metadata to the shared catalogue, SWALLOW.

The SpokenWeb “Repository” currently holds files from Concordia, SFU and uAlberta, and each level of the file structure has controlled access through user passwords and permissions. If members want to search other institution’s holdings, they search the metadata describing files in SWALLOW.

Publishing and archiving your data



Data Flow Model: Sharing and Preservation

Researcher question: Is data curation and data sharing the same thing? Does data curation help me publish peer-reviewed articles or is it only for digital publications like podcasts, websites, and online exhibitions? What are data papers?

Data Flow model questions that help you think through this complexity: Which data requires long-term preservation? Will data be published, or shared with audiences beyond the partnership? Will metadata be shared?

If you go this far in the data curation management flow, then the decision about what to share or archive (preserve) should be clearer. You can follow the FAIR principle of REUSE as a starting point; if you want your data to be reusable, what needs to be shared to ensure your data can be used and understood – and then ensure you also stick to the consent and copyright sections – what needs to be restricted and how. As data curation is a resource-heavy undertaking, another best practice to follow is to determine what you can sustainably and ethically make reusable. This, again, is context-specific to your project, funding, and data set attributes.

Digital humanities projects can produce many kinds of publications across multiple media platforms, including, for example, traditional publications (monograph, article), data sharing, knowledge mobilization through non-traditional digital formats, data papers, and web pages. These digital assets can be archived in a repository as well, alongside the data for the purposes of long-term preservation – but these outputs might live as two different versions in two different places. Archiving data is not the same as preserving other digital research outputs. Further, the working storage solution (that allows RAs and researchers to work, save, share their data while the project is ongoing) and the data preservation solution both have different requirements in terms of rights, formats, capacity, protection, and life span. A solution has to be worked out for both in parallel. Finally, when working in partnerships, sharing and preservation must be planned in a way to recognize contributions, and ensure licenses are followed, as well as publishing and sharing agreements are carried out. Some institutions and infrastructures might already have user agreements in place that treat all research partners/owners equally.

A “circulation copy” or “access copy” of a workshop tutorial video might be published to a website and hosted on a commercial platform such as YouTube to reach the widest audience possible and for user friendly streaming, but this approach does not guarantee long term preservation of the video. For this, you could archive a preservation copy in a repository as “data.” Although in digital humanities circles, the word “repository” can be used as a metaphorical language to describe a custom built and privately hosted digital asset management solution without a long-term preservation plan. Similarly, you could always store data, research work, or other project related outputs that you don’t want to share publicly but want to preserve or share in the short-term using dark storage. In research data management networks, however, a preservation-ready data repository is a specialized software platform built to archive the data for as long as the technological infrastructure is maintained. Some examples of data repositories of interest to digital humanists are listed below; many large-scale DH partnerships are international in nature and so the repository options are global:

- [Zenodo](#) (international in scope)
- [Scholar's Portal Dataverse](#) and [FRDR](#) (available to researchers working in Canada or their international partners, equivalents exist in other national settings)
- [CORE Open Access for the Humanities](#) (international in scope, discipline specific)

These platforms are an option for you to both publish your data for sharing with the public and self-archive your data on a stable preservation ready platform. Submit your final data files to a repository assigning a persistent identifier (e.g., handles or DOIs). Provide good metadata for your study so others can find it (your discipline's metadata standard can be used, e.g., Dublin Core, DDI, etc.). Here again you may want to check back in with your institution's library and archives; they may offer digital preservation services for researchers.

Data papers are a growing genre of computational methodology papers that accompany a published data set. For example, the [Journal of Cultural Analytics](#) is a peer-reviewed venue for this kind of data publication. Additional peer-reviewed venues include the Journal of Open Humanities Data. Not only does it share your data while providing your theoretical and critical framework for collecting and working with your data, it also provides a valuable teachable resource for future generations of Digital Humanities scholars who will learn how to do computational work on carefully curated and well-documented data sets.

Future Directions

Time, space, and labour are limited resources in the research data-verse. We have included this section as an acknowledgement of all the additional topics digital humanists WISH were in this Primer. This is a way of speaking these other resources into being in the future and advocating for the needs of digital humanists in our national data infrastructure and training ecosystem. If you need support with any of the topics below, we recommend you contact your librarian or other specialized research support staff. They are key to the successful undertaking of data curation management.

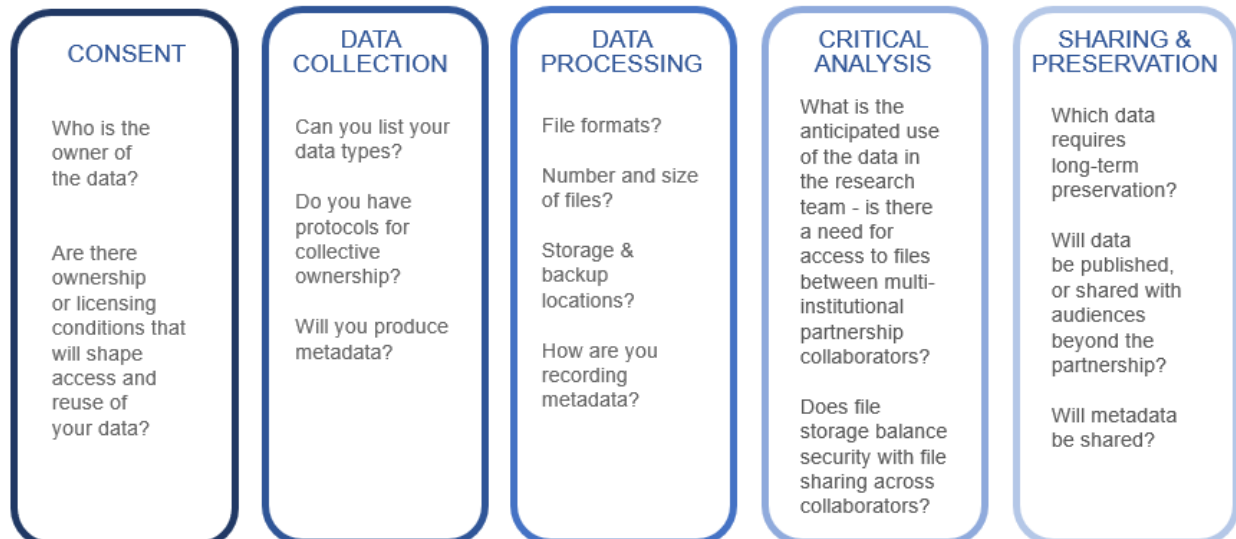
1. Ethical use of AI and Big Data analysis in the Humanities; for example, working with consensual and non-consensual internet data use, with a roles and relationship approach.
2. Working with digital collections that have Indigenous community ownership and protocols for engagement, for example:
 - a. A guide to developing data sharing, use and ownership agreements that go beyond the REB requirements or intellectual property laws.
 - b. Guidelines for third party data stewardship when working with community data ownership protocols (e.g., in support of Indigenous data sovereignty)
3. Guide for artists and creative workers on RDM related topics, for example:
 - a. Creative outputs as data, if you are working inside academic publishing and funding contexts
 - b. Negotiating engagement with academic research projects, when you are working outside of this context, with an emphasis on artists' rights and community representation
4. Humanities-friendly guidance to how RDM can help support disciplinary challenges
 - a. File Storage and best practices for data sharing within research teams
 - b. Best practices for managing data through an equity, diversity, and inclusion lens across multiple institutional partners
 - c. Long term preservation of webpages and other web-based resources
 - d. Interpreting and applying copyright law and open licenses when working across different national environments

- e. How and where to find secondary data
- f. Metadata standards in the Humanities, including metadata schema recommendations and best practices in object description

Annex 1 – Cleaned up worksheet: SpokenWeb Data Flow

Breakout group: Data Flow & Discovery Model

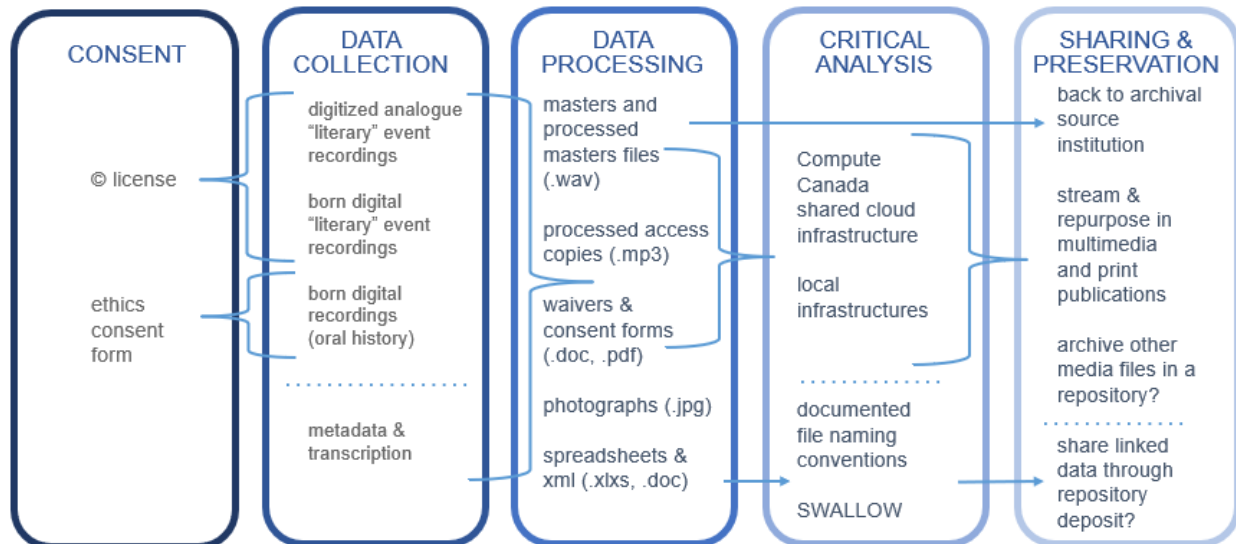
Diagram your data flow through different phases of your project. Discover data you didn't know you had! 📌



Based on Eugene Barsky, "Data Flow and Discovery Manual, Version 11," 28 July 2014, UBC Library, <https://cancoprdrn.wordpress.com/2014/07/28/assessing-researchers-data-needs-research-data-discovery-manual>

Breakout group: Data Flow & Discovery Model

Diagram your data flow through different phases of your project. Discover data you didn't know you had! 📌



Based on Eugene Barsky, "Data Flow and Discovery Manual, Version 11," 28 July 2014, UBC Library, <https://cancoprdrp.wordpress.com/2014/07/28/assessing-researchers-data-needs-research-data-discovery-manual>

Data Flow & Discovery Model

CONSENT	DATA COLLECTION	DATA PROCESSING	CRITICAL ANALYSIS	SHARING & PRESERVATION
Who is the owner of the data?	Can you list your data types?	File formats?	What is the anticipated use of the data in the research team - is there a need for access to files between multi-institutional partnership collaborators?	Which data requires long-term preservation?
Are there ownership or licensing conditions that will shape access and reuse of your data?	Do you have protocols for collective ownership?	Number and size of files?	Does file storage balance security with file sharing across collaborators?	Will data be published, or shared with audiences beyond the partnership?
	Will you produce metadata?	Storage & backup locations?		Will metadata be shared?
		How are you recording metadata?		

The flowchart is organized into five vertical columns representing stages of data handling:

- CONSENT**: Contains a circle with the number '1' and the text 'Consent form'.
- DATA COLLECTION**: Contains three numbered items: '1. Interview recordings (analogue)', '2. Broadcast events recordings', and '3. Social history'. Below these is 'metadata transcription'.
- DATA PROCESSING**: Contains '1. Masters', '2. WAV', '3. acous/voicebank', '4. mp3', '5. digital content', '6. photos', '7. video', and '8. spreadsheets'.
- CRITICAL ANALYSIS**: Contains 'compute', 'cloud', 'university?', 'documentation', 'rich', 'naming', 'conventions', and 'DS wallon'.
- SHARING & PRESERVATION**: Contains 'back to oris collection', 'stream on web', 'journals', 'repository?', 'open linked data?', and 'data?'.

Arrows indicate the flow of data from left to right between these stages. At the bottom, there is a note: 'OPEN LINKED DATA??? HERE TO DEPOSIT FOR PRESERVATION???' and another note on the right: 'More room on back of page'.

21 May 2021

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Making Digital Humanities Research Data Public Data Primer Bibliography

This bibliography is not comprehensive. It is a collection of works which informed our thinking for each section of the data primer that we think could be of interest to you too.

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